

TISM Message Exchange - Messaging packets and exchanges

Version 1.0

Use case:	This document describes a simple data exchange protocol designed to facilitate short messages between microcontrollers ('hosts') that use the TISM programming framework and are connected to the same bus network. The protocol can be implemented as a Stop and Wait ARQ (Automatic Repeat Request) or (small) sliding window version. It is connectionless, non-routed, aimed at exchange of TISM-messages that contain commands (e.g. controlling actuators) or reading sensor values across different network types (like RS485, Wifi, UART or 433Mhz radio). The protocol also contains a number of features to discover other hosts and services ('tasks') in the network. Furthermore, separate logical networks across the same medium are supported using a Network ID. In the TISM-framework a payload is maximum 16 bytes; this version of the protocol supports up to 16 bytes of payload.
Remarks:	Broadcast messages are not acknowledged by receiving hosts. Timeout for responses set to 200 msec for slow connections. Retry limit set to 3. Bus arbitration (e.g. RS485) is not included. Sequence numbers set to 8 bits and intended to wrap around, which should not be an issue in a Stop and Wait ARQ sequence or limited sliding windows. When responding to broadcasts, hosts use their own sequence number for UARTEMX_SEQ_NUMBER. Standard packet sizes range from 12 to 28 bytes. TISM predefined messages have fixed sizes: - TISM basic message and acknowledgement packets: 13 bytes - TISM connectivity, host and task test packets: 16 bytes - TISM host discovery messages: 20 bytes Payload size up to 255 (0xFF) bytes can be addressed by UARTEMX_MESSAGE_SIZE; in this version of the protocol message sizes > 16 (0x1C) are not supported. Maximum packet size is 28 (0x1C) bytes; maximum 16 bytes payload (UARTEMX_MESSAGE_SIZE is 0x10), 11 bytes overhead. TISM_MESSAGE_TYPE determines how a packet (and payload) should be processed by the recipient. Custom/user defined message types range from 0x64 (100) to 0xFF (255). Each host broadcasts a TISM_INTRODUCE packet every 5 minutes; starting with a random delay between 0-60 seconds before the first broadcast (to prevent collisions).

TISM basic message and acknowledgement packets

TISM Message transmission: Sender sends a TISM message to a specific host, or broadcasts to all hosts in the network. For both a TaskID can be specified.

Number of bytes: 12-28

										Payload				
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE TYPE	(9) TISM_ PAYLOAD0	(13) TISM_ PAYLOAD1	(17) TISM_MESSAGE_ TIMESTAMP	(25) UARTMX_ CHECKSUM	(27) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	4 bytes	8 bytes	2 bytes	1 byte
	First header byte													
	0x0 (2)	0x0-0xF (3)	0x00-0xFF (4)	0x00-0x10 (5)	0x01-0xFE (6)	0x00-0xFE (7)	0x01-0xFE (6)	0x00-0xFE (8)	User specified (9)	Optional: Payload0 (10)	Optional: Payload1 (10)	Optional: Timestamp (11)	CRC16 (12)	
0xFF broadcast														

(1) See "UartMX markers & message types" tab for reserved values.

(2) 0x0 indicates UartMX protocol version 1.0.

(3) 4 bit network ID, identifying a group of hosts. Packets with different network IDs are ignored (comparable to VLAN). Default value is 0xF ('everyone').

(4) 8 bit Sequence number set by TISM_UartMX, wraps around.

(5) A standard TISM message packet has 0 (0x00), 4 (0x04), 8 (0x08) or 16 (0x10) bytes payload.

(6) 0x00 is an invalid HostID only to be used on the local host, 0xFF used for broadcasting.

(7) 0xFF is an invalid Sender_TaskID.

(8) If a packet is addressed to a single host and the recipient's TaskID is unknown, set to 0x00. The TISM framework offers a way for tasks to 'subscribe' to custom message types and receive packets in that way.

(9) TISM_MESSAGE_TYPE determines how packet should be processed by recipients. Custom/user-defined values range from 0x64 (100) to 0xFF (255).

(10) Transmission of Payload0 and Payload1 is optional. When omitted, a 0x00 value is added by the receiving host.

(11) Transmission of the sender's timestamp is optional. When omitted, the timestamp is added by the receiving host (timestamp of packet reception) to the TISM message. When used, a value for Payload0 and Payload1 (4 bytes each) is mandatory.

(12) CRC-16-CCITT Bit-by-Bit-Fast used for checksum calculation, with polynomial 0x1021, initial value 0xFFFF, calculated starting with the first header byte.

Packet acknowledgement (ACK)/Negative acknowledgement (NAK): Recipient confirms (acknowledges) correct/incorrect reception of packet.

Number of bytes: 13

										Payload		
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE TYPE	(9) TISM_ PAYLOAD0	(10) UARTMX_ CHECKSUM	(12) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	2 bytes	1 byte
	First header byte											
	0x0	0x0-0xF	0x00-0xFF	0x01 (1)	0x01-0xFE	0x00 (2)	0x01-0xFE	0x00 (3)	TISM_ACK or TISM_NAK (4)	Sender's UARTMX_ SEQ_NUMBER (5)	CRC16	

Note: ACK/NAK packets are only sent when a received packet is NOT a broadcast. An ACK packet is always 13 bytes in size (1 byte payload).

(1) As a ACK/NAK-packet has 1 byte data payload, UARTMX_MESSAGE_SIZE for a ACK/NAK packet is 0x01.

(2) Sender TaskID is always 0x00, as TISM itself sends packets.

(3) Recipient TaskID is always 0x00

(4) Depending on packet type (ACK/NAK); see "UartMX markers & message types" tab.

(5) Sender's second header byte, UARTMX_SEQ_NUMBER, is returned, to identify the packet.

TISM connectivity, host and task test packets

TISM_PING - PING REQUEST: Sender sends a 'ping' request to check the availability of one host in the network, or a specific task (Task ID) running on the specified host. The host or addressed task responds with a TISM_ECHO packet.
Number of bytes: 16

										Payload		
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE TYPE	(9) TISM_ PAYLOAD0	(13) UARTMX_ CHECKSUM	(15) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	2 bytes	1 byte
	First header byte											
	0x0	0x0-0xF	0x00-0xFF	0x04 (1)	0x01-0xFE	0x00-0xFE (2)	0x01-0xFE Host ID (3)	0x00 PingHost (2) 0x01-0xFE PingTask ID (3)	TISM_PING	Identifier payload (4)	CRC16	

Note: TISM_PING packets are not confirmed with an ACK/NAK packet by the recipient.

(1) As a ping-request has a 4 bytes payload, UARTMX_MESSAGE_SIZE is always 0x4.

(2) When sending a ping-request to a Host ID (not a specific TaskID), use 0x00.

(3) 0x00 is an invalid HostID, TISM_PING using broadcast (0xFF) is not allowed, use TISM_DISCOVER instead.

(4) Unique identifier set by sender to identify an echo response; the recipient must respond with the same identifier.

TISM_ECHO - PING REPLY: Recipient replies with an 'echo' message to confirm availability.

Number of bytes: 16

										Payload		
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE TYPE	(9) TISM_ PAYLOAD0	(13) UARTMX_ CHECKSUM	(15) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	2 bytes	1 byte
	First header byte											
	0x0	0x0-0xF	0x00-0xFF	0x04 (1)	0x01-0xFE	0x00-0xFE	Sender's host ID 0x01-0xFE (2)(3)	Sender's task ID 0x00-0xFE (3)	TISM_ECHO	Identifier payload (4)	CRC16	

Note: TISM_ECHO packets are not confirmed with an ACK/NAK packet by the recipient.

(1) As a ping-reply ('echo') has 4 bytes payload, UARTMX_MESSAGE_SIZE is always 0x4.

(2) 0x00 is an invalid HostID and 0xFF for broadcasting. Echo reply broadcast is not allowed.

(3) Ping replies ('echos') are sent directly to the requesting HostID, not a broadcast.

(4) Unique identifier set by sender to identify an echo response; the recipient must reply with the same identifier.

TISM host discovery messages

TISM_INTRODUCE: Sender introducing itself to the network, or respond to a TISM_DISCOVER request from a host. Data includes it's hostname (max. 8 bytes) and number of running tasks. When used to reply to a broadcast, each host applies a random delay up to 20msec to limit the risk of packet collisions. Aside from responding to TISM_DISCOVER packets, each host sends out TISM_INTRODUCE broadcasts periodically.

Number of bytes: 20

										Payload			
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE TYPE	(9) TISM_ PAYLOAD0	(13) TISM_ PAYLOAD1	(17) UARTMX_ CHECKSUM	(19) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	4 bytes	2 bytes	1 byte
	First header byte												
	0x0	0x0-0xF	0x00-0xFF	0x08 (1)	0x01-0xFE	# registered tasks (2)	0xFF (3)	0x00 (4)	TISM_INTRODUCE	First 4 bytes of sender's hostname (5)(6)	Last 4 bytes of sender's hostname (6)	CRC16	

(1) The TISM_INTRODUCE packet has 8 bytes data as payload, UARTMX_MESSAGE_SIZE always 0x8.

(2) Number of registered tasks on the sending host, cannot be 0xFF.

(3) Introduction packet is always a broadcast.

(4) Only the TISM system uses the discover feature. Other values than 0x00 are invalid.

(5) First character of hostname is MSB.

(6) When a hostname is shorter than 8 bytes, the string is padded with 0x00's, all 8 payload bytes are mandatory.

TISM_DISCOVER: Sender sends a 'discover' broadcast to the network, or unicast to a single host, to look for host ID's, hostnames and number of running tasks on these hosts. When a HostID or hostname is provided (in the payload fields), only the host with the corresponding HostID or hostname will respond with a TISM_INTRODUCE broadcast.

Number of bytes: 20

										Payload			
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE_TYPE	(9) TISM_ PAYLOAD0	(13) TISM_ PAYLOAD1	(17) UARTMX_ CHECKSUM	(19) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	4 bytes	2 bytes	1 byte
	First header byte												
	0x0	0x0-0xF	0x00-0xFF	0x08 (1)	0x01-0xFE	0x00 (2)	0x01-0xFE (3) 0xFF Broadcast	0x00 (2)	TISM_DISCOVER	First 4 bytes of recipient's hostname (4)(5)	Last 4 bytes of recipient's hostname (4)(5)	CRC16	

(1) As the 'Discover' packet has 8 bytes data as payload, UARTMX_MESSAGE_SIZE always 0x8.

(2) Only the TISM system uses the discover feature. Other values than 0x00 are invalid.

(3) TISM_DISCOVER can be directed to a single host, or used as a broadcast. When a HostID is provided, payload fields should be empty.

(4) All 8 payload bytes are mandatory. When a hostname is shorter than 8 bytes, the string is padded with 0x00's. When no hostname is provided, payload fields are filled with 0x00's.

(5) When payload is provided, it will be interpreted by the receiving hosts that only the host with the corresponding hostname needs to respond with a TISM_INTRODUCE message.

TISM_RESOLVE_TASKNAME: **Sender** requests one or all hosts for the TaskID of the specified task name.

Number of bytes: 20

										Payload			
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE TYPE	(9) TISM_ PAYLOAD0	(13) TISM_ PAYLOAD1	(17) UARTMX_ CHECKSUM	(19) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	4 bytes	2 bytes	1 byte
	First header byte												
	0x0	0x0-0xF	0x00-0xFF	0x08 (1)	0x01-0xFE	0x00 (3)	0x01-0xFE (2) 0xFF Broadcast	0x00 (3)	TISM_RESOLVE_ TASKNAME	First 4 bytes of task name (4)(5)	Last 4 bytes of task name (5)	CRC16	

Note: When the requested hostname is not found, the recipients of the request return 0xFF for task ID.

(1) As the 'resolve task name' packet has 8 bytes data as payload, UARTMX_MESSAGE_SIZE always 0x8.

(2) 0x00 is an invalid HostID, 0xFF used for for broadcasting.

(3) Only the TISM system uses the resolve feature. Other values than 0x00 are invalid.

(4) First character of taskname is MSB.

(5) When a taskname is shorter than 8 bytes, the string is padded with 0x00's.

TISM_RESOLVE_TASKNAME_REPLY: **Recipient** sends a broadcast reply packet to all hosts, which includes the requested task ID (when found). When a TISM_RESOLVE_TASKNAME message is sent as a reply to a broadcast, NO reply is sent when the TaskID cannot be resolved. Before replying to a broadcast each host applies a random delay up to 20msec to limit the risk of collisions.

Number of bytes: 20

										Payload			
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE TYPE	(9) TISM_ PAYLOAD0	(13) TISM_ PAYLOAD1	(17) UARTMX_ CHECKSUM	(19) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	4 bytes	2 bytes	1 byte
	First header byte												
	0x0	0x0-0xF	0x00-0xFF	0x08 (1)	0x01-0xFE	0x00-0xFE Requested Task ID	0xFF (3)	0x00	TISM_RESOLVE_ TASKNAME_REPLY	First 4 bytes of requested task name (4)(5)(6)	Last 4 bytes of requested task name (4)(6)	CRC16	
						0xFF Requested Task name not found (2)							

(1) The 'resolve task ID reply' packet has 8 bytes as payload, so it is always 0x8.

(2) When the requested taskname cannot be resolved, TISM_SENTER_TASKID is 0xFF (TaskID 0x00 is usually for the TISM-scheduler).

(3) The TISM_RESOLVE_TASKNAME_REPLY packet is always a broadcast, so all hosts can update their records.

(4) The task name provided by the sender is returned, to allow matching of this reply to the original request.

(5) First character of taskname is MSB.

(6) When a taskname is shorter than 8 bytes, the string is padded with 0x00's. When the requested TaskID cannot be resolved, payload is filled with 0x00 only.

TISM_RESOLVE_TASKID: **Sender** requests one or all hosts (broadcast) for the task name of the specified TaskID, or all tasks running on one specific host.

Number of bytes: 20

										Payload			
(0) UARTMX START_MARKER	(1) UARTMX VERSION_ID	(1) UARTMX NETWORK_ID	(2) UARTMX SEQ_NUMBER	(3) UARTMX MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGETYPE	(9) TISM_ PAYLOAD0	(13) TISM_ PAYLOAD1	(17) UARTMX_ CHECKSUM	(19) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	4 bytes	2 bytes	1 byte
	First header byte												
	0x0	0x0-0xF	0x00-0xFF	0x08 (1)	0x01-0xFE	0x00	0x01-0xFE (2)	0x00	TISM_RESOLVE_ TASKID	RequestedTaskID 0x01-0xFE (3)	Not used (5)	CRC16	
						0xFF Broadcast	Request all tasks 0xFF (3)(4)						

(1) The TISM_RESOLVE_TASKID packet has 8 bytes payload, UARTMX_MESSAGE_SIZE=0x08.

(2) 0x00 is an invalid HostID, 0xFF used for broadcasting.

(3) TaskID of the task to be resolved; the value 0xFF is a request to a host to resolve all existing tasks on that host. TaskID is stored in the 4th byte/LSBs.

(4) A value of 0xFF is NOT allowed when TISM_RECIPIENT_HOSTID=0xFF (broadcast), to prevent flooding of the network.

(5) Reserved/unused, filled with padding(0x00).

TISM_RESOLVE_TASKID_REPLY: **Recipient** sends a broadcast including the requested task name, so all hosts can update their records. When replying to a broadcast, a response is ONLY sent when the requested TaskID can be resolved. Before replying, each host applies a random delay up to 20msec to limit the risk of collisions.

Number of bytes: 20

										Payload			
(0) UARTMX_ START_MARKER	(1) UARTMX_ VERSION_ID	(1) UARTMX_ NETWORK_ID	(2) UARTMX_ SEQ_NUMBER	(3) UARTMX_ MESSAGE_SIZE	(4) TISM_ SENDER_HOSTID	(5) TISM_ SENDER_TASKID	(6) TISM_ RECIPIENT_ HOSTID	(7) TISM_ RECIPIENT_ TASKID	(8) TISM_ MESSAGE_TYPE	(9) TISM_ PAYLOAD0	(13) TISM_ PAYLOAD1	(17) UARTMX_ CHECKSUM	(19) UARTMX_ END_MARKER
1 byte	4 bits	4 bits	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	4 bytes	4 bytes	2 bytes	1 byte
	First header byte												
	0x0	0x0-0xF	0x00-0xFF	0x08 (1)	0x01-0xFE	RequestedTaskID 0x00-0xFE (2)	0xFF	0x00	TISM_RESOLVE_ TASKID_REPLY	First 4 bytes of task name (3)(4)	Last 4 bytes of task name (3)	CRC16	

(1) The 'resolve task ID reply' packet has 8 bytes as payload, so it is always 0x08.

(2) TISM_SENDER_TASKID contains the resolved TaskID. This is an exception to the normal field usage where TISM_SENDER_TASKID identifies the sending task. When the TaskID cannot be resolved, 0x00 for TISM_SENDER_TASKID is returned. Value cannot be 0xFF.

(3) When the requested TaskID cannot be resolved, payload is filled with 0x00 only. When a taskname is shorter than 8 bytes, the string is padded with 0x00's.

(4) First character of taskname is MSB.

TISM Message Exchange - Protocol Markers and Message Types

TISM_Messaging structure:

Field name	Size	Description
SenderHostID	1 byte	Host ID of the sender of the message
SenderTaskID	1 byte	Task ID of the task sending the message
RecipientHostID	1 byte	Host ID of the recipient
RecipientTaskID	1 byte	Task ID of the recipient
MessageType	1 byte	Type of message; see below or user defined
Payload0	4 bytes	Payload
Payload1	4 bytes	Payload
MessageTimestamp	8 bytes	Number of usec since power on of host.

Packet markers:

Marker	Value	Description
UARTMX_START_MARKER	0xD3	Start of packet marker
UARTMX_END_MARKER	0x55	End of packet marker

UARTMX_VERSION_ID values:

Value	Comment
0x00	UartMX Standard Version

Other values reserved for future use. Packets with different version IDs are ignored.

Message types:

Used in the TISM messaging system. TISM system message type values range from 50 (0x32) to 99 (0x63).

Message types below are specifically used for message exchange between hosts.

Message Type	Value	Packet size	UARTMX MESSAGE_SIZE	Description
TISM_TEST	0x32	20	0x08	Dummy message, does nothing.
TISM_ACK	0x33	13	0x01	Positive acknowledgement of message receipt
TISM_NAK	0x34	13	0x01	Negative acknowledgement of message receipt
TISM_PING	0x35	16	0x04	Check to see if recipient host or task is still functional.
TISM_ECHO	0x36	16	0x04	Response to ping-request.
TISM_DISCOVER	0x37	20	0x08	Broadcast; discovery of available hosts, hostnames and running tasks.
TISM_INTRODUCE	0x38	20	0x08	Host introducing itself to the network.
TISM_RESOLVE_TASKNAME	0x39	20	0x08	Request for TaskID belonging to specified task name
TISM_RESOLVE_TASKNAME_REPLY	0x3A	20	0x08	Reply; requested TaskID
TISM_RESOLVE_TASKID	0x3B	20	0x08	Request for TaskName belonging to specified task ID
TISM_RESOLVE_TASKID_REPLY	0x3C	20	0x08	Reply; requested TaskName